

# CISCO SECURE WORKLOAD

## HIPAA Compliance Framework

Prepared for: [Customer Name]

Prepared by: Cisco Systems, Inc.

Date: [Month Year] | Version 1.0 | Cisco Confidential

**Regulatory Standard: HIPAA Security Rule (45 CFR Part 164)**

Environment: Hybrid (On-Premises + AWS / Azure / GCP)  
Deployment Stage: Fresh Installation

## How to read the Coverage column

**Direct** means Secure Workload produces the primary evidence artifact for that row — segmentation, policy, flow and process telemetry, inventory, vulnerability and reachability, or forensic reconstruction. Direct is still evidence, not an attestation.

**Supporting** means Secure Workload contributes an input that feeds or complements another control, tool, or process — detection input only, reconciled against an external register, or one log source among several.

**Evidence Required** means Secure Workload has no record for that row. The customer supplies the contract, the analysis, or the HR / physical-security evidence from another control.

Every report in this set now uses the same three coverage words — Direct, Supporting, and Evidence Required. Coverage describes what evidence Secure Workload produces; it never means the control is passed.

## Executive Summary

Healthcare organizations face escalating regulatory obligations under the HIPAA Security Rule to protect electronic Protected Health Information (ePHI). This report documents how Cisco Secure Workload (CSW) enables [Customer Name] to satisfy HIPAA Security Rule requirements through workload-level visibility, micro-segmentation, real-time monitoring, and forensic telemetry.

CSW provides the technical foundation for a defense-in-depth approach to ePHI protection — ensuring that access to sensitive clinical and billing data is continuously authorized, monitored, and auditable across on-premises and cloud environments.

## Compliance Posture Summary

| Safeguard Category                   | Controls Addressed                                                                    | Coverage          |
|--------------------------------------|---------------------------------------------------------------------------------------|-------------------|
| Administrative Safeguards §164.308   | Risk analysis, access management, incident response, contingency planning, evaluation | Direct            |
| Physical Safeguards §164.310         | Workload isolation and access boundary enforcement                                    | Supporting        |
| Technical Safeguards §164.312        | Access controls, audit controls, integrity, authentication, encryption in transit     | Direct            |
| Organizational Requirements §164.314 | Business Associate scope definition, third-party access control                       | Evidence Required |

*Note: Physical Safeguards relate to physical facility controls (locks, cameras, workstation policies) which are outside CSW scope. CSW directly complements these controls through workload isolation and logical access enforcement.*

# CSW Capabilities & HIPAA Alignment

Cisco Secure Workload delivers four core capability pillars that directly address HIPAA Security Rule requirements:

| CSW Capability                       | What It Does                                                                                                                                  | HIPAA Controls Satisfied                                                                              |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Application Dependency Mapping (ADM) | Discovers all workload-to-workload communication flows, processes, and users. Identifies ePHI application boundaries without prior knowledge. | §164.308(a)(1) Risk Analysis<br>§164.312(a) Access Control<br>§164.308(a)(3) Workforce Access         |
| Micro-Segmentation                   | Enforces allowlist policies at the workload level — not just perimeter firewalls. Isolates ePHI systems from unauthorized lateral access.     | §164.312(a)(1) Access Control<br>§164.312(e) Encryption/Isolation<br>§164.308(a)(6) Incident Response |
| Vulnerability Detection              | Continuously scans workloads for CVEs with CVSS scoring. Prioritizes risk remediation on ePHI-bearing systems.                                | §164.308(a)(1) Risk Analysis<br>§164.308(a)(8) Evaluation                                             |
| Forensic Telemetry                   | Retains network flow and process-level audit trail across all monitored workloads for breach investigation and compliance reporting.          | §164.312(b) Audit Controls<br>§164.308(a)(6) Incident Response<br>§164.308(a)(1) Risk Analysis        |

## HIPAA Control Mapping Detail

The following table maps each applicable HIPAA Security Rule control to the corresponding CSW capability and implementation approach.

| Safeguard      | Ref                | Control                           | HIPAA Requirement                                                                               | CSW Implementation                                                                                                                                | Status |
|----------------|--------------------|-----------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Technical      | §164.312(a)(2)(i)  | <b>Unique User Identification</b> | Assign unique name or number for identifying and tracking user identity.                        | CSW ADM captures process-user context per workload. Anomaly detection alerts on shared or generic credential usage patterns.                      | Direct |
| Technical      | §164.312(a)(2)(ii) | <b>Emergency Access Procedure</b> | Establish procedures for obtaining necessary ePHI during an emergency.                          | Break-glass policy workspace configured in CSW with alert-only mode. Emergency IP ranges defined with expanded but audited access.                | Direct |
| Technical      | §164.312(b)        | <b>Audit Controls</b>             | Implement mechanisms to record and examine activity in systems containing ePHI.                 | CSW retains full network flow and process telemetry. Flow search enables retrospective audit of any ePHI workload interaction.                    | Direct |
| Technical      | §164.312(c)(1)     | <b>Integrity Controls</b>         | Protect ePHI from improper alteration or destruction.                                           | CSW policy enforcement blocks unauthorized write-path access. Process hash verification detects workload tampering.                               | Direct |
| Technical      | §164.312(d)        | <b>Authentication</b>             | Verify that a person or entity seeking access to ePHI is the one claimed.                       | CSW enforces LDAPS/Kerberos-only paths to ePHI systems. Unencrypted LDAP (port 389) blocked via policy.                                           | Direct |
| Technical      | §164.312(e)(2)(ii) | <b>Encryption in Transit</b>      | Implement a mechanism to encrypt ePHI in transit when appropriate.                              | CSW detects unencrypted protocols (HTTP, FTP, plain LDAP) on PHI-Zone workloads. Policy enforces HTTPS/SFTP/LDAPS.                                | Direct |
| Administrative | §164.308(a)(1)     | <b>Risk Analysis</b>              | Conduct an accurate and thorough assessment of the potential risks and vulnerabilities to ePHI. | CSW vulnerability detection provides continuous CVE assessment on all PHI-Zone workloads. CVSS-scored reports exportable for risk register input. | Direct |

|                |                |                                     |                                                                                                                |                                                                                                                                                   |                   |
|----------------|----------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Administrative | §164.308(a)(3) | <b>Workforce Access Management</b>  | Implement procedures for authorizing access to ePHI.                                                           | CSW scope-based policies define which workloads can access ePHI systems. Policy changes require approved workflow.                                | Direct            |
| Administrative | §164.308(a)(6) | <b>Security Incident Procedures</b> | Implement policies to address security incidents, including identifying and responding to suspected incidents. | CSW anomaly detection and policy violation alerts provide real-time incident detection. Forensic telemetry supports full incident reconstruction. | Direct            |
| Administrative | §164.308(a)(7) | <b>Contingency Plan</b>             | Establish policies for responding to emergencies that damage systems containing ePHI.                          | CSW policy workspaces exportable for DR purposes. Backup policy sets maintained for recovery scenarios.                                           | Direct            |
| Administrative | §164.308(a)(8) | <b>Evaluation</b>                   | Perform periodic technical and nontechnical evaluation of implemented security measures.                       | CSW compliance dashboard provides ongoing posture evaluation. Quarterly ADM re-runs detect policy drift.                                          | Direct            |
| Organizational | §164.314(a)    | <b>Business Associate Contracts</b> | Document satisfactory assurances from business associates that they will appropriately safeguard ePHI.         | CSW defines External-Partners scope with restricted policy. All BA network access flows through audited, policy-enforced paths.                   | Evidence Required |

# Recommended Deployment Architecture

## PHI Zone Segmentation Model

CSW organizes workloads into policy scopes that map directly to HIPAA data classification requirements. The following architecture is recommended for hybrid healthcare environments:

| PHI Zone (Strict)                                                                                       | Clinical Support (Moderate)                                               | Corporate IT (Standard)                                                    | External Partners (BAA)                                              |
|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------|
| EHR / EMR Servers<br>Billing Systems<br>Medical Imaging (PACS)<br>HL7 Integration<br>Clinical Databases | Clinical Workstations<br>Lab Systems<br>Pharmacy Apps<br>Diagnostic Tools | Active Directory<br>Email / Collaboration<br>File Servers<br>IT Management | Clearinghouses<br>Third-Party Apps<br>Health Plans<br>Vendor Portals |
| Default DENY all inbound<br>Encrypted paths only<br>Full forensic logging                               | Restricted to defined PHI-Zone flows<br>ADM-validated policies            | No direct PHI access<br>Jump-host controlled admin paths                   | BAA-documented access only<br>Full flow audit per partner            |

## Hybrid Environment Coverage

CSW provides unified policy and visibility across on-premises and cloud workloads:

| Environment | CSW Deployment Method                                                                     | Capabilities                                                                    |
|-------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| On-Premises | Software agent (Linux/Windows sensor)<br>Direct CSW cluster registration                  | Full telemetry, policy enforcement, vulnerability scan, process context         |
| AWS         | Software agent on EC2<br>Cloud connector (VPC Flow Logs)<br>Tag-based scope inheritance   | Flow visibility, security group mapping, instance inventory, policy enforcement |
| Azure       | Software agent on VMs<br>Cloud connector (NSG Flow Logs)<br>Azure tag synchronization     | Flow visibility, NSG mapping, VM inventory, policy enforcement                  |
| GCP         | Software agent on GCE<br>Cloud connector (VPC Flow Logs)<br>Label-based scope inheritance | Flow visibility, firewall rule mapping, instance inventory                      |

## Evidence Collection & Audit Readiness

CSW generates and retains the evidence required to demonstrate HIPAA compliance to auditors. The following evidence artifacts are available from CSW:

| Evidence Artifact                 | CSW Source                                  | HIPAA Control  | Recommended Frequency         |
|-----------------------------------|---------------------------------------------|----------------|-------------------------------|
| Policy Enforcement Log            | <i>Defend &gt; Policy Analysis</i>          | §164.312(a)    | Continuous / Monthly export   |
| Vulnerability Scan Report         | <i>Investigate &gt; Vulnerability</i>       | §164.308(a)(1) | Weekly                        |
| Network Flow Audit Log (PHI-Zone) | <i>Investigate &gt; Flow Search</i>         | §164.312(b)    | Continuous / Quarterly export |
| Policy Violation Report           | <i>Alerts &gt; Triggered Events</i>         | §164.308(a)(6) | Real-time / Monthly summary   |
| Scope Membership Snapshot         | <i>Inventory &gt; Export</i>                | §164.308(a)(3) | Monthly                       |
| Encryption Compliance Report      | <i>Flow analysis (protocol filter)</i>      | §164.312(e)    | Quarterly                     |
| Application Dependency Map        | <i>Investigate &gt; ADM</i>                 | §164.308(a)(1) | Quarterly re-run              |
| Forensic Incident Timeline        | <i>Investigate &gt; Flow/Process Search</i> | §164.308(a)(6) | On-demand (post-incident)     |

## Documentation Retention

HIPAA requires documentation to be retained for a minimum of 6 years from the date of creation or the date it was last in effect. CSW supports this through:

- Configurable telemetry retention periods (recommend 6-year minimum for audit logs)
- Policy workspace version history — tracks all policy changes with timestamps
- Exportable evidence packages in PDF/CSV format for long-term archival
- API-based integration with SIEM/SOAR platforms for centralized log management

## Recommended Next Steps

| # | Phase                                 | Action                                                                                                        | Timeline   |
|---|---------------------------------------|---------------------------------------------------------------------------------------------------------------|------------|
| 1 | <b>Sensor Deployment</b>              | Install CSW software sensors on all on-prem PHI workloads. Connect cloud accounts via CSW cloud connectors.   | Days 1-5   |
| 2 | <b>Scope &amp; Inventory Design</b>   | Create PHI-Zone scope hierarchy. Apply HIPAA labels (env, compliance, data) across all workloads.             | Days 6-10  |
| 3 | <b>Application Dependency Mapping</b> | Run 2-4 week ADM to discover all ePHI application flows. Identify and tag ePHI apps from ADM output.          | Days 11-28 |
| 4 | <b>Policy Development</b>             | Build allowlist policies from ADM baseline. Configure default-deny for PHI-Zone. Run in Simulation mode.      | Days 29-42 |
| 5 | <b>Enforcement Go-Live</b>            | Progress from Simulation to Enforcement. Validate no clinical workflow disruption. Enable full alerting.      | Days 43-56 |
| 6 | <b>Compliance Reporting</b>           | Generate first HIPAA evidence package. Schedule quarterly review cycle. Deliver compliance dashboard to CISO. | Day 60+    |

### About Cisco Secure Workload

Cisco Secure Workload (CSW) is a workload protection platform that provides application micro-segmentation, real-time vulnerability assessment, and forensic telemetry across hybrid and multi-cloud environments. CSW is deployed by leading healthcare, financial services, and critical infrastructure organizations to satisfy regulatory compliance requirements and accelerate zero trust adoption.

For more information or to schedule a technical deep-dive, contact your Cisco account team.